

Experiment No. 8:

To determine the internal resistance of a cell using potentiometer.

8.1 Objectives

The main objective of this lab is to measure the internal resistance of a given primary cell with the help of a potentiometer.

8.2 Prelab

Student must have clear idea about the objectives and related theory of the experiment.

8.3 Outcomes

After completing this lab work, students will be able to answer the following questions:

- What is a potentiometer? Why is it called potentiometer?
- What is the principle of a potentiometer?
- What is internal resistance?

8.4 Timing and Length of Investigation (Total 3 Hours)

- **Lab Preparation (15 minutes):**
 - For attendance and equipment distribution among the groups.
- **Lecture on Theory (30 minutes):**
 - Teacher will clarify the objective and theory of the experiment.
- **Lecture on Procedure (15 minutes):**
 - Demonstrator will make the students understand the procedure of the experiment.
- **Experimental Work (90 to 100 minutes):**
 - Design of experiment: 10 minutes.
 - Data collection: 30 minutes.
 - Data analysis, graph drawing and completing the result part: 50 to 60 minutes.
- **Post Lab Discussion (15 to 20 minutes):**
 - Teacher will summarize the total lab work and have a discussion with the students related with the questions given in the outcomes part.
- **Report Submission:**
 - Student will prepare their lab reports as groups in MS word within one week.
 - Students must submit their group lab reports as printed before the starting of the next lab and give viva.

8.5 Theory

A cell or any other source which supplies a potential difference to the circuit to which it is connected has within it some resistance called internal resistance. When there is no current in the cell *i.e.*, at open circuit, the potential difference E between its terminal is maximum and is called its *electro-motive force (e. m. f.)*. When the cell is discharging *i.e.*, at closed circuit, its terminal potential difference is reduced to e because of the internal drop of potential across its internal resistance b .

In Fig. 8.1 the balance point for the cell E_1 whose internal resistance b is to be determined, is found out as usual, at a distance l_1 from the end A of the potentiometer with key K_2 open. Then a resistance R is introduced in the resistance box RB and the key K_2 is closed. The potential difference between the terminals of the cell E_1 falls as a current i begins to flow through the circuit. A balance point is now found at a distance l_2 from the end A of the potentiometer. As E and e are the potential difference at the open and closed circuits and b is the

internal resistance of the cell. We have $b = \frac{E-e}{i}$ where i is the current flowing through the circuit when the key K_2 is closed. Again $i = \frac{e}{R}$

$$b = \frac{E-e}{e} R = \left(\frac{E}{e} - 1 \right) R \dots \dots \dots (1)$$

But as E and e are proportional to l_1 and l_2 , we have $\frac{E}{e} = \frac{l_1}{l_2}$

from (1),

$$b = \left(\frac{l_1}{l_2} - 1 \right) R = \frac{l_1 - l_2}{l_2} R \dots \dots \dots (2)$$

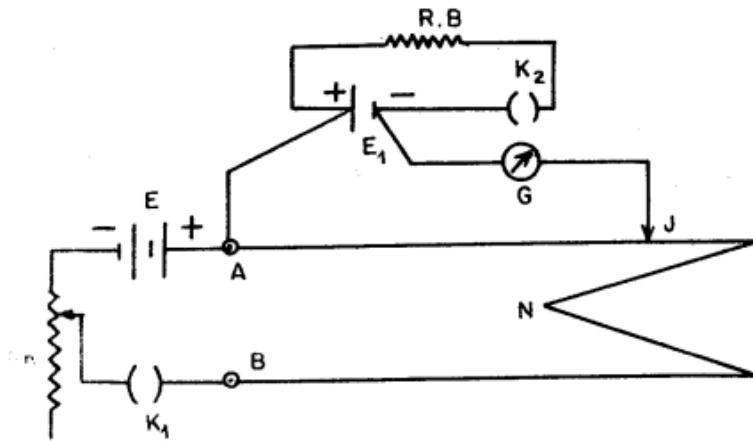


Figure 8.1: Circuit for to determine the internal resistance of a cell using potentiometer

8.6 Apparatus

Potentiometer, battery E , cell E_1 , resistance box R , rheostat R_h , two keys K_1 and K_2 , zero-centre galvanometer G , connecting wires.

8.7 Procedure

- Connect the positive terminal of the battery E to the binding screw A of the potentiometer and the negative terminal of the battery through R_h and the key K_1 to the binding screw B of the potentiometer (Fig 8.1). Join the positive terminal of the cell E_1 whose internal resistance is to be determined, to the binding screw A of the potentiometer and its negative terminal through the galvanometer G to the jockey J . Also connect the resistance box $R.B$ through the key K_2 to the two terminals of the cell E_1 . It is better to put a shunt across the galvanometer.
- Adjust a small resistance in the rheostat R_h and close the key K_1 . Keep the key K_2 open and press the jockey first near the end A and then near the end B of the potentiometer wire. If the galvanometer deflections are in the same direction, then either the resistance in R_h is too great or $e.m.f$ of E is too small. Decrease the resistance in R_h until the opposite deflections are obtained at the above two contact points. If necessary, increase the number of cells in the battery E . Adjustment of R_h should be such as to get a null point on the fifth or sixth wire.

- Remove the shunt of the galvanometer (if any) and find out the balance point accurately. Open the key K_1 and calculate the distance l_1 of the balance point from the end A of the potentiometer wire.
- Close the key K_2 without changing R_h and take out a resistance 10 ohms from the R.B and determine the balance point and calculate the distance l_2 of the balance point from A. Then remove 20, 30, 40, 50 ohms from R.B and determine the value of l_2 in each case.
- Calculate the value of b from the relation (2) for each value of R and then calculate the mean value of b .

8.8 Experimental Data

Experimental Data:
Table: Balance point on the Potentiometer.

No. of Obs.	Circuit	Resistance ohms	Value of				Internal Resistance b ohms	Mean b ohms	
			l_1 cm	Mean l_1 cm	l_2 cm	Mean l_2 cm			
	Open	Infinity			×	×	×		
1	Closed	10	×						
2	Closed	20	×						
3	Closed	30	×						
4	Closed	40	×						
5	Closed	50	×						

8.9 Calculation:

Internal resistance $b = \left(\frac{l_1}{l_2} - 1\right)R$

(1) -----

(2) -----

8.10 Result:

Internal resistance = -----ohm

8.11 Discussion

8.12 Resources

For further understanding students may go through the following resources:

- **Fundamental of Physics (10th Edition):** Potential difference (Chapter 27, page 778-788).
- **Video Links:**
 - [Prof. Hemant Dongre, Internal Resistance of Cell using Potentiometer - YouTube](#)